

Geometric Quantization in Mathematical Quantum Physics

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Hamiltonian Mechanics on Euclidean Spaces

In classic mechanics, two factors comes in to play when determining the motion of a particle with mass m in $\mathbb{R}^n$

- The position vector $\mathbf{x} = (x^0, x^1, \dots, x^{n-1})$ and momentum $\mathbf{p} = (p^0, p^1, \dots, p^{n-1})$
- The energy (**Hamiltonian**) of the particle given its position $\mathbf{x}$ and momentum $\mathbf{p}$. This is a function (often smooth) $H(\mathbf{x}, \mathbf{p})$ over the **phase space** $\mathbb{R}^{2n}$

They tells us the trajectory of the particle, starting at $(\mathbf{x}_0, \mathbf{p}_0)$, through the following Hamilton's equation

$$\frac{d}{dt}x^i(t) = \frac{\partial}{\partial p^i}H, \frac{d}{dt}p^j(t) = -\frac{\partial}{\partial x^j}H,$$

Newtonian Mechanics

If we let

$$H(\mathbf{x}, \mathbf{p}) = \frac{|\mathbf{p}|^2}{2m} + V(\mathbf{x})$$

The Hamilton's equation then becomes

$$\frac{d}{dt}x^i(t) = \frac{p^i}{m}, \quad \frac{d}{dt}p^i(t) = -\frac{\partial}{\partial x^i}V(\mathbf{x})$$

If we let $F = -\nabla V(\mathbf{x})$, then we have Newton's Second Law

$$F = m \frac{d^2}{dt^2} \mathbf{x}(t)$$

Poisson Bracket

Definition

For smooth functions $f, g : \mathbb{R}^{2n} \rightarrow \mathbb{R}$, the **Poisson bracket** is

$$\{f, g\} = \frac{\partial f}{\partial x^i} \frac{\partial g}{\partial p^i} - \frac{\partial f}{\partial p^i} \frac{\partial g}{\partial x^i}$$

The Poisson bracket determines a Lie algebra structure on $C^\infty(\mathbb{R}^{2n})$.

Conserved Quantity

Given a Hamiltonian H , for any smooth function $f : \mathbb{R}^{2n} \rightarrow \mathbb{R}$ and any $\gamma(t)$ that satisfy the Hamilton's equation,

$$\frac{d}{dt}f(\gamma(t)) = \{f, H\}(\gamma(t))$$

A smooth function f is conserved if $f'(\gamma(t)) = 0$ for all $\gamma(t)$, in other words

$$\{f, H\} = 0$$

Hamiltonian Mechanics on Symplectic Manifold

We wish to generalize the concepts of Hamiltonian mechanics over Euclidean space to a more general manifolds.

Definition

A **symplectic manifold** (M, ω) contains a smooth manifold M and a 2-form ω such that

- $d\omega = 0$
- for any $p \in M$, $\omega_p^\flat : T_p M \rightarrow T_p^* M$ defined as $\omega_p^\flat(X)(Y) = \omega_p(X, Y)$ is a linear isomorphism.

A **symplectomorphism** $f : (M, \omega_M) \rightarrow (N, \omega_N)$ is a smooth map, such that $f^* \omega_N = \omega_M$.

Hamiltonian Vector Field

Definition

Given $f \in C^\infty(M)$, the **Hamiltonian vector field** is the unique vector field X_f such that $df = \omega^\flat(X_f)$.

Example

An example would be $\mathbb{R}^{2n}$ with $\omega = dp^i \wedge dx^i$, the Hamiltonian vector field of x^i and p^i are

$$X_{x^i} = \frac{\partial}{\partial p^i}, X_{p^i} = -\frac{\partial}{\partial x^i}$$

Moreover, $\omega(X_f, X_g) = -\{f, g\}$

Since nondegenerate bilinear $\omega : V \otimes V \rightarrow \mathbb{R}$ that is antisymmetric exists only for $\dim V = 2n$, we have that symplectic manifold must be even dimension.

Hamiltonian System

Definition

For $f, g \in C^\infty(M)$, the Poisson bracket is $\{f, g\} := -\omega(X_f, X_g)$.

(Warning: There are different sign convention for different text, we follow [1] as it is our main reference. However, it is positive in [3])

Definition

A **Hamiltonian System** (M, ω, H) consists of a symplectic manifold (M, ω) and a smooth function H , the Hamiltonian. The **Hamiltonian flow**, Φ_H is the integral flow of the Hamiltonian vector field $-X_H$.

Time-Evolution along Hamiltonian Flow

Similar to the Euclidean case, we have the following property.

Theorem

Given a smooth function f , we have that for every $p \in M$

$$\frac{d}{dt}f(\Phi_H(t)) = \{f, H\}(\Phi_H(t))$$

Proof.

We have $\frac{d}{dt}f(\Phi_H(t)) = -X_H(f) = -\omega(X_f, X_H) = \{f, H\}$ □

Symplectic Structure on Cotangent Bundle

Example

Let M be any smooth manifold, its cotangent bundle $\pi : T^*M \rightarrow M$, there exists one particular 1-form θ defined for each vector field X at each point $(p, \alpha_p) \in T_p^*M$

$$\theta(X)(p, \alpha_p) := \alpha_p(\pi_*(X))$$

There is then a symplectic form $\omega = d\theta$ over T^*M .

It is not hard to see for the case $M = \mathbb{R}^n$, $\theta(\mathbf{x}, p_i dx_i) = p_i dx_i$ and $d\theta = dp_i \wedge dx_i$. Precisely the symplectic form for Euclidean space.

Basics of Quantum Physics

Quantum Physics studies structures of some Hilbert space $\mathcal{H}$.

- A **state** is a density operator ρ of $\mathcal{H}$.
- An **observable** is a (unbounded) self-adjoint operator $\mathcal{O}$ on $\mathcal{H}$.
- Let $\sigma(\mathcal{O})$ be the spectrum of $\mathcal{O}$, there exists a **projection valued measure** $\mu^{\mathcal{O}}$ due to Spectral theorem. Define

$$\text{Prob}(\{\text{observation of } \mathcal{O} \text{ at state } \rho\} \in E \subseteq \sigma(\mathcal{O})) = \text{tr}(\rho \mu^{\mathcal{O}}(E))$$

The expected value of $\mathcal{O}$ at ρ is then

$$E_{\rho}[\mathcal{O}] = \int_{\lambda \in \sigma(\mathcal{O})} \lambda \text{tr}(\rho \mu^{\mathcal{O}}(d\lambda)) = \text{tr}(\rho \int_{\lambda \in \sigma(\mathcal{O})} \lambda \mu^{\mathcal{O}}(d\lambda)) = \text{tr}(\rho \mathcal{O})$$

Time-Evolution

Now we fix an observable $\hat{H}$, call it the **Hamiltonian**, which determines time-evolution like in classical world.

Definition (Schrodinger's equation in the Heisenberg Picture)

An observable $\mathcal{O}(t)$ where $\mathcal{O}(0) = \mathcal{O}$ evolves according to

$$\frac{d}{dt}\mathcal{O}(t) = \frac{-i}{\hbar}[\mathcal{O}(t), \hat{H}]$$

Here $[\cdot, \cdot]$ is the commutator bracket.

This is solved by $\mathcal{O}(t) = U(t)^* \mathcal{O} U(t)$, here $U(t) := \exp(-i\frac{\hat{H}}{\hbar}t)$ is called the **time-evolution operator**.

From Classic to Quantum

Suppose we have a symplectic manifold (M, ω) , one often want

Physical Quantity f on $M \rightarrow$ Observable $\hat{f}$ on $\mathcal{H}$

There is then question

How to **Quantize** Physical Quantities on a Symplectic Manifold?

We need a **quantization scheme** that takes good functions on M to operators on some $\mathcal{H}$

Quantization

Definition

A **prequantization** consist of a Hilbert space $\mathcal{H}$ and a linear transformation

$$Q : \mathcal{A} \subseteq C^\infty(M) \otimes \mathbb{C} \rightarrow \mathcal{L}(\mathcal{H})$$

Here $\mathcal{A}$ is a Lie subalgebra (closed under $\{\cdot, \cdot\}$ containing 1. Q satisfies

- $Q(1) = I$
- $Q(\bar{f}) = Q(f)^*$
- $i\hbar Q(\{f, g\}) = [Q(f), Q(g)]$ because $-\frac{i}{\hbar}[\cdot, \cdot]$ and $\{\cdot, \cdot\}$ have the same role in time-evolution

If moreover Q satisfies some irreducibility condition on the observables, it is called a **quantization**. [2]

Weyl Quantization

Let $M = \mathbb{R}^{2n}$, **Weyl quantization** defined for $\mathcal{A} = \mathbb{C}[x_i, p_i]_{deg \leq 2}$ over $\mathcal{H} = \mathcal{L}^2(\mathbb{R}^n)$ is a commonly used quantization, where

$$Q_{\text{Weyl}}(x_i) = X_i := x_i, \quad Q_{\text{Weyl}}(p_i) = P_i := -i\hbar \frac{\partial}{\partial x_i}$$

And some multiplicative properties.

This cannot be made better.

Theorem (Groenewold's Theorem)

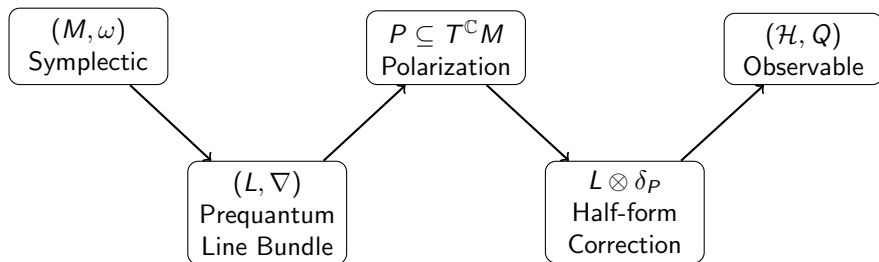
Let $\mathcal{D}(\mathbb{R}^n)$ be the differential operators of polynomial coefficient over $\mathbb{R}^n$, there does not exist a linear transformation $Q : \mathbb{C}[x_i, p_i]_{deg \leq 4} \rightarrow \mathcal{D}(\mathbb{R}^n)$ such that

- ① Q is a quantization, where $Q(x_i) = X_i, Q(p_i) = P_i$
- ② Image lies in polynomials of X_i, P_i

Geometric Quantization for Symplectic Manifolds

Geometric quantization is a daring project aiming to construct a quantization for symplectic manifolds (M, ω) .

Geometric quantization consists of



Connections

Let $\pi : E \rightarrow M$ be a vector bundle.

Definition

A **connection** is a bilinear transformation $\nabla : TM \otimes E \rightarrow E$

$$\begin{aligned}\nabla_{fX}(s) &= f\nabla_X(s), \\ \nabla_X(fs) &= f\nabla_X(s) + X(f)s\end{aligned}$$

For $X \in TM$, $s \in E$, and f smooth.

Connections are differentiation of sections along vector field [4].

Example (Jacobian)

Let $E = T\mathbb{R}^n$, for $X, Y \in TM$

$$\nabla_X Y := (JY)X$$

where JY is the Jacobian, determines a connection of $T\mathbb{R}^n$

Parallel Transport

Definition

For any smooth curve $\gamma : [0, 1] \rightarrow M$, denote $\dot{\gamma}(t) := \gamma_*(d/dt)$, a section $s \in E|_{\gamma}$ is **parallel** respect to γ if

$$\nabla_{\dot{\gamma}} s = 0$$

The equation above reduce to a set of first order PDE's. Therefore, given $s(0)$, $s(t)$ is unique

Specifically, a linear transformation $P_{\gamma} : E|_{\gamma(0)} \rightarrow E|_{\gamma(1)}$, $s(0) \mapsto s(1)$, the **parallel transport** of ∇ along γ .

If γ is a loop, so $\gamma(0) = \gamma(1)$, the **holonomy** of ∇ along γ , $\text{Hol}_{\gamma}^{\nabla} := P_{\gamma}$, which is an automorphism of $E|_{\gamma(0)}$.

Hermitian Connection

Definition

A **Hermitian line bundle** is a complex line bundle $\pi : L \rightarrow M$ with a smooth pairing of sections $(\cdot, \cdot)$ that is an inner product on each fiber. A **Hermitian connection** is a connection ∇ on L such that,

$$(s_1, \nabla_X s_2) + (\nabla_X s_1, s_2) = X(s_1, s_2)$$

Definition

The **curvature form** of a Hermitian connection ∇ is a 2-form Ω_∇ , such that for any $X, Y \in TM$

$$\Omega_\nabla(X, Y)s = i(\nabla_X \nabla_Y - \nabla_Y \nabla_X - \nabla_{[X, Y]})s$$

Curvature Form of a Hermitian Connection

For $U \subseteq M$ where L is trivializable, there exists $s \in L|_U$ such that $(s, s) = 1$. Since

$$(s, \nabla_X s) + (\nabla_X s, s) = 0, X \in TU$$

$\nabla_X s = -i\theta(X)s$, where $\theta \in T^*M$. As it turns out $\Omega_\nabla = d\theta$ over U .

Given a loop $\gamma(t)$, solving the equation

$$\nabla_{\dot{\gamma}}(fs) = 0$$

gives us

$$\text{Hol}_\gamma^\nabla = \exp(i \int_\gamma \theta)$$

Integral 2-Form

If γ lies on the boundary of some 2-d submanifold S

$$\mathrm{Hol}_{\partial S}^{\nabla} = \exp(i \int_S \Omega_{\nabla})$$

And if S is closed (compact and no boundary), $\mathrm{Hol}_{\partial S}^{\nabla} = 1$. This is equivalent to $\int_S \Omega_{\nabla} \in 2\pi\mathbb{Z}$.

We call a 2-form ω **integral** if for all closed 2-d submanifold S , $\int_S \omega \in \mathbb{Z}$.

Theorem

For every closed 2-form ω such that $\omega/2\pi$ is integral, there exists a Hermitian connection (L, ∇) over M such that $\Omega_{\nabla} = \omega$.

See [2] for the construction of the line bundle.

Geometric Prequantization

Definition

A symplectic manifold (M, ω) is **quantizable** if $\omega/2\pi\hbar$ is integral.

Let (L, ∇) be a choice of Hermitian connection corresponding to $\omega/\hbar$.

Definition

The **prequantum Hilbert space** $\mathcal{H}_{\text{pre}}$ is the space of sections $s \in L$ such that

$$\|s\|^2 := \int_M (s, s) \text{vol}_M < \infty$$

Where $\text{vol}_M := \omega^{\wedge n}/n!$. The **geometric prequantization** of f is

$$Q_{\text{pre}}(f) := i\hbar \nabla_{X_f} + f$$

This is indeed a prequantization of $C^\infty(M)$.

Harmonic Oscillator

Example

Let $M = \mathbb{R}^2$, and let $H = \frac{1}{2m}p^2 + \frac{1}{2}m\omega^2x^2$. We have

$$Q_{\text{pre}}(p^2) = i\hbar[m\omega^2x\frac{\partial}{\partial p} - \frac{p}{m}\frac{\partial}{\partial x}]$$

This operator has eigenvalues $n\hbar\omega$ for $n \in \mathbb{Z}$.

However, the quantum mechanic harmonic oscillator should have eigenvalue $\hbar\omega(n + \frac{1}{2})$. Something is wrong.

Polarization

From the harmonic oscillator, the geometric prequantization is not the right quantization.

In Weyl quantization, we look at the L^2 functions only respect to the position. These are functions $\nabla_{X_p}s = 0$.

We might fix a set of direction of the manifold and look at the sections that are constant along this direction.

Definition

Given a symplectic manifold (M, ω) of dimension $2n$, a **polarization** is a subbundle $P \subseteq T^{\mathbb{C}}M := TM \otimes \mathbb{C}$ such that

- $\dim P = n$ and $\omega(X, Y) = 0$ for all $X, Y \in P$
- if $X, Y \in P$, then $[X, Y] \in P$.

Purely Real of Purely Complex

Given a polarization $P \subseteq T^{\mathbb{C}}M$, we call P **purely real** if $P = \bar{P}$, and we call it **purely complex** if $P \cap \bar{P} = \{0\}$.

In this talk,

Polarization are Purely Real.

However, purely complex polarization is very interesting as well. Let P be purely complex, there is a decomposition of $T^{\mathbb{C}}M = P \oplus \bar{P}$, which can be given an almost complex structure, and take our discussion to the field of complex geometry.

Vertical Polarization of Cotangent Bundle

Example

Let M be a manifold, the cotangent bundle T^*M admits a polarization, the **vertical polarization**, such that $P|_{T_p^*M} = T(T_p^*M)$.

It is the subbundle of $T(T^*M)$ spanned by the tangent vector $\frac{\partial}{\partial p^i}$ along the direction of the cotangent fiber.

One attempt to Geometric Quantization

Replicating the Euclidean case, we might let $\mathcal{H} \subseteq \mathcal{H}_{\text{pre}}$ contains sections s such that for all $X \in \mathcal{P}$

$$\nabla_X s = 0$$

We call s a **polarized section**.

This however, fails to be nontrivial for many cases.

Example

Vertical polarization on T^*M have $\text{Hol}^\nabla = 1$ on the leaves. The polarized sections are the ones that are constant along the fiber, yet the integral of sections constant along the fiber are ∞ unless $s = 0$.

However, the set of leaves $\Xi \cong M$ in this case is also a manifold, and we can restrict the norm to be integrated over Ξ instead of T^*M .

We look at elements in form $s \otimes \sqrt{dx}$, where s is polarized, and $\sqrt{dx}$ is a “square root” of the volume form on Ξ , so $\|s \otimes \sqrt{dx}\|^2 = \int_{\Xi} (s, s) dx$ here (s, s) is viewed as a function depends only on x .

Example

In Weyl quantization, our Hilbert space is $\mathcal{L}^2(\mathbb{R}^n)$, the functions that only depends on $\mathbf{x}$ and are $\mathcal{L}^2$

Half Form

Definition

Given a polarization P , the **canonical bundle** K_P is the real line subbundle of $\bigwedge^n T^*M$ such that for any $X \in P$

$$\iota_X \alpha = 0$$

Moreover, α is **polarized** if $\iota_X d\alpha = 0$ as well. This is equivalent as saying $\mathcal{L}_X \alpha = 0$.

Let $q : M \rightarrow \Xi$ sending a point to the leaf it belongs to, equip Ξ with the weak topology induced by q . If Ξ is a smooth manifold, then

Theorem

[1] *For every polarized section α there is a unique section $\tilde{\alpha}$ over Ξ such that $\alpha = q^* \tilde{\alpha}$ and for any n -form β of Ξ , $q^* \beta$ is polarized.*

If moreover, Ξ admits a nowhere vanishing n -form β . Let $\alpha := q^*\beta$ be a polarized section of M , this is nowhere vanishing so K_P is trivial.

A **square root** of K_P is a line bundle δ_P and a bundle isomorphism $\delta_P \otimes \delta_P \rightarrow K_P$.

We define a connection ∇ on K_P , by $\nabla_X \alpha = \iota_X d\alpha$, since $\iota_X \alpha = 0$, this equals to $\mathcal{L}_X \alpha$. We can define an unique operator δ_P , denoting $\mathcal{L}_X$ as well by defining them on the tensor product, for $s \in \delta_P$

$$\mathcal{L}_X(fs) = X(f)s + f\mathcal{L}_X s, \quad \mathcal{L}_X(s_1 \otimes s_2) = (\mathcal{L}_X s_1) \otimes s_2 + s_1 \otimes (\mathcal{L}_X s_2)$$

For any sections $s_i = \mu_i \otimes \nu_i \in L \otimes \delta_P^{\mathbb{C}}$, here $\delta_P^{\mathbb{C}}$ is the complexification, we define

$$(s_1, s_2) := (\mu_1, \mu_2) \nu_1 \otimes \nu_2 \in K_P^{\mathbb{C}}$$

We call a section $s = \mu \otimes \nu$ **polarized** if

$$\nabla_X s = (\nabla_X \mu) \otimes \nu + \mu \otimes \mathcal{L}_X \nu = 0$$

Given polarized sections s_1, s_2 , (s_1, s_2) is then a polarized section of K_P , hence has a corresponding form (s_1, s_2) on Ξ

Geometric Quantization with Half Form

Definition

Given a quantizable symplectic manifold (L, ∇, M, ω) , a polarization P such that the leaves Ξ is a orientable manifold, and a square root $\delta_P \otimes \delta_P \rightarrow K_P$. The **half form space** $\mathcal{H}$ is the Hilbert space of polarized sections of $L \otimes \delta_P^{\mathbb{C}}$, such that

$$\|s\|^2 = \int_{\Xi} (s, \tilde{s}) < \infty, \quad \langle s_1, s_2 \rangle := \int_{\Xi} (s_1, \tilde{s}_2)$$

The **Geometric Quantization** of a function f such that $[X_f, X] \in P$ for all $X \in P$ is

$$Q(f)s = (Q_{\text{pre}}\mu) \otimes \nu + i\hbar\mu \otimes (\mathcal{L}_{X_f}\nu)$$

Comparing Different Polarization: BKS Construction

Given two polarization, P_1 and P_2 , their intersection $P_1 \cap P_2 := D_{12} \subseteq T_M$ determines leaf space M/D_{12} . Suppose there are half-form δ_1 and δ_2 and

$$\mu_i : \delta_i \otimes \delta_i \rightarrow K_{P_i}$$

There exists $\delta_1 \otimes \delta_2 \rightarrow T^*(M/D_{12})$. We define pairing for $s_i = \mu_i \otimes \nu_i \in L \otimes \delta_i^{\mathbb{C}}$

$$(s_1, s_2)_{12} = \int_{M/D_{12}} (\mu_1, \mu_2)_L (\bar{\nu}_1 \otimes \nu_2)$$

This pairing induces an unique operator $\Lambda_{P_1, P_2} : \mathcal{H}_1 \rightarrow \mathcal{H}_2$ such that

$$(s_1, s_2)_{P_1} = (\Lambda_{P_1, P_2} s_1, s_2)_{P_2}$$

This is called the **BKS (Blattner-Kostant-Sternberg) construction**

General Geometric Quantization

Fix a polarization P and let f be arbitrary such that X_f does not necessarily preserves P . For any $t \in [0, 1]$, the flow $\Phi_t^f : M \rightarrow M$, defines us a new polarization

$$(P_t)_p := (\Phi_t^f)_* P_{(\Phi_t^f)^{-1}(p)}$$

The pullback map $(\Phi_{-t}^f)^* : T^*M \rightarrow T^*M$ takes $K_P \rightarrow K_{P_t}$, and it induces a linear map $(\Phi_{-t}^f)^*$ on the half-form δ_t of P_t .

The prequantum operator $Q_{\text{pre}}(f)$ defines an operator $\exp(\frac{i}{\hbar} t Q_{\text{pre}}(f))$ on $\mathcal{L}$, then there is

$$(\Phi_t^f)^\sharp(\mu \otimes \nu) = \left[\exp(\frac{i}{\hbar} t Q_{\text{pre}}(f)) \mu \right] \otimes \left[(\Phi_{-t}^f)^* \nu \right]$$

From $\mathcal{H} \rightarrow \mathcal{H}_t$

In many cases (for instance the case for vertical polarization of cotangent bundle), Λ_{P,P_t} and $(\Phi_t^f)^\sharp$ are unitary. Therefore, there is an one-parameter unitary group

$$\mathcal{U}^f(t) := \Lambda_{P,P_t}^{-1} \circ (\Phi_t^f)^\sharp$$

Definition

The most general formula for the geometric quantization is

$$Q(f) = -i\hbar \frac{d}{dt} \mathcal{U}^f(t) \Big|_{t=0}$$

For the case X_f preserves P , hence Λ_{P,P_t} is the identity, and the above equation agrees with the previous definition of geometric quantization (using product rule-like argument)

Cotangent Bundle of Riemannian Manifold

Suppose M is a Riemannian manifold. Let $\omega := d\theta$ be the canonical symplectic form, therefore $L \cong M \times \mathbb{C}$.

Let P be the vertical polarization. The leaves $\Xi \cong M$, which has a nonvanishing n -form such that for any basis $e_i \in T_p M$

$$\text{vol}_g(e_1, \dots, e_n) = \sqrt{\det(g(e_i, e_j))}$$

vol_g then determines the polarized form $\pi^* \text{vol}_g$ on T^*M .

Choose half-form δ_P to be trivial, with section $\sqrt{\pi^* \text{vol}_g} \in \delta_P$, and let $\delta_P \otimes \delta_P \rightarrow K_P$ takes $\sqrt{\pi^* \text{vol}_g}^{\otimes 2} \rightarrow \pi^* \text{vol}_g$.

Since $\mathcal{L}_X \pi^* \text{vol}_g = 0$ for all $X \in P$, we have $L \otimes \delta_P^{\mathbb{C}} \cong C^\infty(M) \otimes \mathbb{C}$. The Hilbert space is $\mathcal{L}^2(M, d\text{vol}_g)$.

Quantization of Energy on Riemannian Manifold

Let g^* be the dual metric of M , and let

$$H(\mathbf{x}, \mathbf{p}) = \frac{g^*(p, p)}{2m} + V(\mathbf{x})$$

Be the energy function on the cotangent bundle T^*M .

The geometric quantization of H is then

$$Q(H) = -\frac{\hbar^2}{2m}(\Delta - \frac{1}{6}R(\mathbf{x})) + V(\mathbf{x})$$

Here Δ is the Laplace-Beltrami operator and R is the scalar Riemann curvature of M [5].

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